AMD Ryzen 7 9700X

A Performance Powerhouse

₹35,999

Performance 8.3
Value for money...7.8

The AMD Zen 5 launch has been quite an interesting one. It's bringing with it a new architecture, it's built on a newer process node and AMD's made some interesting choices with regards to the kind of workloads that the new AMD Ryzen 9000 processors would be optimised for. In this review, we're looking at the AMD Ryzen 7 9700X that's priced at approximately INR 35,999 in India. The Ryzen 7 9700X is positioned slightly lower in the hierarchy, making it more accessible for users who want top-tier performance but don't need as many cores as the Ryzen 9. With 8 cores and 16 threads, this chip is aimed at gamers, creators, and general PC enthusiasts who want a balance between performance and power consumption. What remains to be seen is if this really sports a decent bump in performance over the Ryzen 7 7700X that was launched two years ago.

SPECIFICATIONS

The 9700X clocks at a base frequency of 3.8 GHz and boosts up to 5.5 GHz, offering strong single-core and multi-core performance. The L3 cache is substantial, with 32 MB, ensuring good gaming and productivity workloads. Unlike the Ryzen 7 8700G, the 9700X doesn't sport the new integrated graphics capabilities, its iGPU is more in line with what launched with the older Ryzen 7000 series processors. So you can certainly play all the usual eSports games on the processor but for everything else, you'll need to pair this with a decent discrete processor. This aligns it more with performance-driven builds rather than budget or hybrid builds.

Shifting from the TSMC N5 to N4P comes with a 22 per cent improvement in power efficiency and when you factor in the reduced base clock frequency from 4.5 GHz to 3.8 GHz, we can see why the Ryzen 7 9700X runs at a TDP of 65 Watts. So what does that mean for the performance? It will be more or less similar to the Ryzen 7 7700 with a little improvement in certain benchmarks. Considering that fact that memory stability on DDR5 has improved quite a bit in these last two years, we have a lot more faster memory sticks in the market and using a slightly faster memory will allow for quick and easy gains.

PERFORMANCE

On paper, the Ryzen 7 9700X positions itself as a solid mid-to-high-tier option for most modern gaming and productivity workloads. During Cinebench multi-threaded runs, the 9700X is competitive, scoring just below the competing Intel Core i5-14600K and but compared to the older gen Ryzen 7 7700X, we saw it score decently higher in the single-threaded run. In multi-threaded runs, the 9700X doesn't have that much of a gain over the 7700X. Switching to a higher memory frequency i.e. 6000 MT/s vs 5600 MT/s saw the 9700X widen the performance gap over the 7700X. This was quite the surprise. We were expecting much bigger gains but that's what you get by reducing the base clock speeds. In y-cruncher, the 9700X has a pretty decent improvement over the 7700X in multi-threaded runs. In Blender, we see decent gains over the 7700X but the 14600K beats both the processors.

In Productivity benchmarks, the Ryzen 7 9700X boasts of meagre gains over the 7700X in UL Procyon Office benchmark for Microsoft Outlook and PowerPoint. The gains are much